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Estimating Number of Factors by Adjusted Eigenvalues Thresholding

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ABSTRACT

Determining the number of common factors is an important and practical topic in high-dimensional factor models. The existing literature is mainly based on the eigenvalues of the covariance matrix. Owing to the incomparability of the eigenvalues of the covariance matrix caused by the heterogeneous scales of the observed variables, it is not easy to find an accurate relationship between these eigenvalues and the number of common factors. To overcome this limitation, we appeal to the correlation matrix and demonstrate, surprisingly, that the number of eigenvalues greater than 1 of the population correlation matrix is the same as the number of common factors under certain mild conditions. To use such a relationship, we study random matrix theory based on the sample correlation matrix to correct biases in estimating the top eigenvalues and to take into account of estimation errors in eigenvalue estimation. Thus, we propose a tuning-free scale-invariant adjusted correlation thresholding (ACT) method for determining the number of common factors in high-dimensional factor models, taking into account the sampling variabilities and biases of top sample eigenvalues. We also establish the optimality of the proposed ACT method in terms of minimal signal strength and the optimal threshold. Simulation studies lend further support to our proposed method and show that our estimator outperforms competing methods in most test cases. Supplementary materials for this article are available online.

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Adjusted eigenvalues; Bias corrections; Factor models; Number of factors; Random matrices

1. Introduction

High-dimensional factor models have many applications in finance, economics, and genomics, or, more generally, in high-dimensional data where the dependence of measurements can be attributed to a relatively small number of common factors (Fan et al. 2020). Determining the number of factors is an important issue in applications of factor models. Methods are typically based on eigenvalues or the rank of the loading matrix. For example, Lewbel (1991) and Kong, Liu, and Zhou (2019) obtained the number of factors by testing the rank of the loading matrix.

Factor models decompose observable multivariate vector $\mathbf{y}$ into latent factors and idiosyncratic components as follows:

$$\mathbf{y} = \boldsymbol{\alpha} + \mathbf{B}\mathbf{f} + \boldsymbol{\epsilon}, \tag{1}$$

where $\mathbf{y} = (y_1, \ldots, y_p)^T$ is the p-dimensional observable vector, $\mathbf{f} = (f_1, \ldots, f_K)^T$ is the K-dimensional latent factor vector, $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_p)^T$ is the p-dimensional error vector which is uncorrelated to the latent factor, $\boldsymbol{\alpha}$ is the p-dimensional intercept vector, and $\mathbf{B}$ is the $p \times K$-dimensional loading matrix. Assuming without loss of generality $\mathrm{cov}(\mathbf{f}) = \mathbf{I}_K$, where $\mathbf{I}_K$ is the $K \times K$ identity matrix, factor model (1) implies that the covariance matrix of $\mathbf{y}$ is

$$\boldsymbol{\Sigma} = (\sigma_{ij}) = \mathbf{B}\mathbf{B}^T + \boldsymbol{\Psi}, \tag{2}$$

where $\mathrm{cov}(\boldsymbol{\epsilon}) = \boldsymbol{\Psi}$. Following Bai and Ng (2002), we define the number of factors as $\mathrm{rank}(\mathbf{B})$, which will be assumed to be K throughout the article. Our goal is to derive a tuning-free and scale-invariant method for selecting K via random matrix theory.

There is a rich source of literature on eigenvalue-based methods for selecting the number of common factors, which have been studied from three different perspectives. The first perspective is model selection. Bai and Ng (2002) proposed three PC and three IC criteria, using penalties to determine the number of common factors. Hallin and Liška (2007) developed an information criterion to determine the number of common factors in a general dynamic model. Li, Li, and Shi (2017) proposed information criteria similar to Bai and Ng (2002) to determine the number of common factors when the number of factors increases with the sample size. Su and Wang (2017) utilized the BIC information criterion to determine the number of common factors for time-varying factor models.

The second perspective involves hypothesis testing or confidence intervals. Onatski (2009) proposed a test statistic $T_{\mathrm{ON}} = \max_{r_{\min} < i \le r_{\max}} (\hat{\lambda}_i - \hat{\lambda}_{i+1})/(\hat{\lambda}_{i+1} - \hat{\lambda}_{i+2})$ to test the number of common factors, where $\hat{\lambda}_i$ is the ith largest eigenvalue of the estimated covariance matrix, and $r_{\min}$ and $r_{\max}$ are the prespecified lower and upper bounds of the number of common factors, respectively. He estimated the number of factors by

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$$\hat{K}_{\mathrm{ON}} = \arg \max_{r_{\min} < i \le r_{\max}} (\hat{\lambda}_i - \hat{\lambda}_{i+1})/(\hat{\lambda}_{i+1} - \hat{\lambda}_{i+2}).$$

Kapetanios (2010) used the statistic $\tau_i(\hat{\lambda}_i - \hat{\lambda}_{r_{\max}+1})$, where τ_i is some normalization constant, to test the number of common factors. Pan and Yao (2008) used the Ljung–Box–Pierce portmanteau test statistic to determine the number of common factors. Based on the confidence interval of the largest nonspiked eigenvalue of the estimated covariance matrix, Cai et al. (2017) proposed an algorithm to determine the number of common factors under a convergence regime that the dimension and the sample size tend proportionally to infinity.

The third perspective is based on estimation. Onatski (2010) used the maximum eigengap to determine the number of common factors and proposed the following eigenvalue difference criterion:

$$\hat{K}_{\mathrm{ED}} = \max\{i \le r_{\max} : \hat{\lambda}_i - \hat{\lambda}_{i+1} \ge t_0\},$$

where t_0 is a given threshold. Wang (2012) proposed using the ratios of two adjacent eigenvalues to determine the number of factors, estimating K by

$$\hat{K}_{\mathrm{ER}} = \arg \max_{1 \le i \le r_{\max}} \hat{\lambda}_i / \hat{\lambda}_{i+1}. \tag{3}$$

Ahn and Horenstein (2013) also independently proposed the "ER" method, in addition to the "GR" method:

$$\hat{K}_{\mathrm{GR}} = \arg \max_{1 \le i \le r_{\max}} \log(V_{i-1}/V_i)/\log(V_i/V_{i+1}),$$

with $V_i = \sum_{j=i+1}^{p} \hat{\lambda}_j$. Lam and Yao (2012) also proposed an ratio estimator to identify the number of factors for time series models by lagged autocovariance matrices, which is further improved by Li, Wang, and Yao (2017) for high-dimensional time series models. Recently, Dobriban and Owen (2019) proposed to select the number of factors by a deterministic parallel analysis and Dobriban (2019) investigated the consistency of traditional parallel analysis which uses random permutations.

The aforementioned methods are all based on the eigenvalues of the covariance matrix. A drawback of the covariance-based methods is that it is not scale invariant and can be inconsistent. For example, taking $\mathbf{\Psi} = \mathrm{diag}(\nu_1^2, \ldots, \nu_p^2)$ in (2), we have

$$\mathbf{\Sigma} = \mathbf{B}\mathbf{B}^T + \mathrm{diag}(\nu_1^2, \ldots, \nu_p^2). \tag{4}$$

Even for the simple factor model with $\mathbf{\Sigma} = \mathbf{B}\mathbf{B}^T + \mathrm{diag}(\underbrace{1, \ldots, 1}_{K}, \nu_{K+1}^2, 1, \ldots, 1)$ where $\mathbf{B}^T = (\mathbf{B}_1^T, \mathbf{0}_{K\times(p-K)})$, $\mathbf{B}_1$ is of $K \times K$-dimension and $\mathrm{rank}(\mathbf{B}_1) = K$, under some mild conditions of $\mathbf{B}_1$, ν_{K+1}^2 and the population distribution, we can show that

$$\begin{aligned} P(\hat{K}_{\mathrm{ON}} \ge K+1) &\to 1, \quad P(\hat{K}_{\mathrm{ED}} \ge K+1) \to 1, \\ P(\hat{K}_{\mathrm{ER}} \ge K+1) &\to 1, \quad P(\hat{K}_{\mathrm{GR}} \ge K+1) \to 1, \end{aligned} \tag{5}$$

but in fact, the true number of common factors is K. The proof of (5) will be given in the supplementary materials, Section S.2.

The correlation matrix clearly overcomes the scaling drawback of the covariance matrix. The $p \times p$-dimensional correlation matrix of $\mathbf{\Sigma}$ is given by

$$\mathbf{R} = [\mathrm{diag}(\mathbf{\Sigma})]^{-1/2} \mathbf{\Sigma} [\mathrm{diag}(\mathbf{\Sigma})]^{-1/2}, \tag{6}$$

where $\mathrm{diag}(\mathbf{\Sigma})$ is the diagonal matrix obtained by replacing the off-diagonal elements of $\mathbf{\Sigma}$ with zeros. In fact, when the dimension is fixed and the sample size tends to infinity, Guttman (1954), Kaiser (1960, 1961), and Johnson and Wichern (2007, p. 491) have shown that the number of eigenvalues satisfying $\max\{j : \hat{\lambda}_j > 1, j \in \{1, \ldots, p\}\}$, where $\hat{\lambda}_j$ is the jth largest eigenvalue of the sample correlation matrix, is no larger than the number of common factors. However, the equality has not yet been established. Moreover, their estimation techniques

$$\hat{K}_u = \max\{j : \hat{\lambda}_j > 1, j \in [p]\}, \quad \text{where} \quad [p] = \{1, \ldots, p\},$$

cannot be consistent in a high-dimensional setting, as the sample correlation matrices are inconsistent.

Can such a simple, tuning parameter-free scale-invariant method be modified so that it is consistent for high-dimensional factor models? This article gives an affirmative answer via some high-dimensional adjustments of threshold parameters and by leveraging random matrix theory.

This article makes three main contributions. First, we establish the concise relationship between the eigenvalues of population correlation matrices and the number of common factors, that is, we give the condition under which

$$K = \max\{j : \lambda_j(\mathbf{R}) > 1, j \in [p]\}, \tag{7}$$

where $\lambda_1(\mathbf{R}) \ge \lambda_2(\mathbf{R}) \ge \cdots \ge \lambda_p(\mathbf{R})$ are the eigenvalues of correlation matrix $\mathbf{R}$ and K is the true number of common factors. As $\sum_{j=1}^{p} \lambda_j(\mathbf{R}) = p$, some eigenvalues of $\mathbf{R}$ are greater than 1 and the rest are no larger than 1. It has been shown (Guttman 1954; Kaiser 1960, 1961) that $K \ge \max\{j : \lambda_j(\mathbf{R}) > 1, j \in [p]\}$. We will show that the equality holds under certain conditions. The results presented in Table 1 illustrate this point when $\{b_{\ell j}, \ell \in [p], j \in [K-1]\}$ are independent and identically distributed (iid) from the uniform distribution $\mathrm{U}(-1, 1)$, $\nu_1^2 = \cdots = \nu_p^2 = \sigma^2$, $b_{1K}, \ldots, b_{pK}$ are iid from $U(-1, 1)$ in Scenario 1 and $b_{1K} = \cdots = b_{pK} = 0$ in Scenario 2. Second, we propose a bias-corrected estimator $\hat{\lambda}_i^C$ for $\lambda_i(R)$, which in general differs from the ith largest eigenvalue $\hat{\lambda}_i$ of sample correlation matrix, and develop a new estimator for the number of common factors as follows:

$$\hat{K}^C = \max\{j : \hat{\lambda}_j^C > s\}, \qquad s = 1 + \sqrt{p/(n-1)} \tag{8}$$

under the regime $\rho_{n-1} = p/(n-1) \to \rho \in (0, \infty)$, where n is the sample size. Our newly proposed method $\hat{K}^C$ does not depend on any tuning parameter and is even simpler than the eigenvalue ratio or difference method $\hat{K}_{\mathrm{ER}}$ or $\hat{K}_{\mathrm{ED}}$, which involves the tuning parameter t_0. On the other hand, by (7), a naive method is

$$\hat{K}_u = \max\{j : \hat{\lambda}_j > 1, j \in [p]\},$$

Table 1. Number of eigenvalues of $\mathbf{R}$ satisfying $\lambda_j(\mathbf{R}) > 1$.

		Scenario 1			Scenario 2		
		$\sigma^2 = 1$	2	3	$\sigma^2 = 1$	2	3
K	p	$\mathrm{rank}(\mathbf{B}) = K$			$\mathrm{rank}(\mathbf{B}) = K - 1$		
5	50	5	5	5	4	4	4
	100	5	5	5	4	4	4
10	50	10	10	10	9	9	9
	100	10	10	10	9	9	9

but this method overestimates K when n and p are of the same order. Let $p/(n-1) \to \rho$ and

$$\mathcal{F}(\nu_0) = \{\mathbf{R}: \ \mathbf{R} \text{ is the correlation matrix of the observed vector in the factor model (1) and } \lambda_K(\mathbf{R}) > \nu_0\},$$

where ν_0 is a positive constant, representing signal strength. We will show that the optimal lower bounds for the signal strength ν_0 and threshold s are

$$\nu_0 = 1 + \sqrt{\rho}, \quad s = 1 + \sqrt{p/(n-1)}$$

in the following sense.

Minimum signal strength ν_0: We will show the following.

(i) When $\nu_0 < 1 + \sqrt{\rho}$, there exists $\mathbf{R} \in \mathcal{F}(\nu_0)$ such that no method based on the eigenvalues of the sample correlation matrix can give a consistent estimate of K.

(ii) When $\nu_0 = 1 + \sqrt{\rho}$, our method $\hat{K}^C$ can consistently estimate K.

Optimal threshold s. Let $\hat{K}^C(s)$ emphasize the dependence of $\hat{K}^C$ on a general s. We will prove that

$$\begin{cases} \text{if } s = 1 + \sqrt{p/(n-1)}, & \forall\, \mathbf{R} \in \mathcal{F}(\nu_0), \quad P(\hat{K}^C(s) = K) \to 1, \\ \text{if } s < 1 + \sqrt{p/(n-1)} - \delta, & \exists\, \mathbf{R} \in \mathcal{F}(\nu_0), \quad P(\hat{K}^C(s) > K) \to 1, \\ \text{if } s > 1 + \sqrt{p/(n-1)} + \delta, & \exists\, \mathbf{R} \in \mathcal{F}(\nu_0), \quad P(\hat{K}^C(s) < K) \to 1. \end{cases}$$

In other words, the threshold $s = 1+\sqrt{p/(n-1)}$ is optimal. We have conducted extensive simulations to compare our method with those in Bai and Ng (2002), Onatski (2005, 2010), Lam and Yao (2012), Ahn and Horenstein (2013), and Li, Wang, and Yao (2017). The simulation results show that in most of our test cases, our estimation method outperforms the competing ones. Even in the remaining cases considered in this article, our estimation method has comparable performance to that of competing methods. Third, we derive the asymptotic properties of the largest K sample eigenvalues of the sample correlation matrix in a high-dimensional factor model. This is an important contribution to random matrix theory. The results may be useful for other inference problems in high-dimensional factor models.

The organization of this article is as follows. Section 2 reviews the factor model and establishes the relationship between the number of common factors and the eigenvalues of the population correlation matrix. Section 3 proposes an estimator of the number of common factors based on a study on random matrix theory of a sample correlation matrix and demonstrates the convergence of the proposed estimator for the number of common factors. Section 4 investigates the optimality of the proposed estimator in high-dimensional factor models. Section 5 presents extensive simulation results. Section 6 conducts two empirical studies. Section 7 concludes the article. Most technical proofs are given in the supplementary materials.

2. High-Dimensional Factor Model

Let us consider factor model (1) that satisfies the following conditions.

- Condition C1: The factors $f_1, \ldots, f_K$ are mutually independent; the factor vector $(f_1, \ldots, f_K)$ is independent of the error vector $(\epsilon_1, \ldots, \epsilon_p)$.
- Condition C2: $\mathrm{E}(\mathbf{f}) = \mathbf{0}_K, \mathrm{cov}(\mathbf{f}) = \mathbf{I}_K$.
- Condition C3: $\mathrm{E}(\boldsymbol{\epsilon}) = \mathbf{0}_p, \mathrm{cov}(\boldsymbol{\epsilon}) = \boldsymbol{\Psi} > \mathbf{0}_{p\times p}$ where $\boldsymbol{\Psi}$ is diagonal or more generally satisfies the sparsity condition $m_p = \max_{i\le p} \sum_{j\le p} |\sigma_{ij}|^q = o(p)$ for some $q \in [0, 1]$.
- Condition C4: $p > K$ and the loading matrix $\mathbf{B}$ is of full column rank.

Write $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_K)$, where $\mathbf{b}_j = (b_{1j}, \ldots, b_{pj})^T$ is a p-dimensional column vector for $j \in [K]$. If there is at most one coefficient $b_{\ell j} \neq 0$ with $\ell \in [p]$ for some $j \in [K]$, that is, f_j is only related to y_j, we can put f_j in ϵ_j. This leads us to the following definition and condition.

Definition of common factors: If there are at least two coefficients $b_{\ell_1 j}, b_{\ell_2 j} \neq 0$ with $\ell_1, \ell_2 \in [p]$ for some $j \in [K]$, we call the factor f_j a common factor.

Condition C5: For every $j \in [K]$, there are at least two coefficients $b_{\ell_1 j}, b_{\ell_2 j} \neq 0$ with $\ell_1, \ell_2 \in [p]$.

Conditions C1 and C2 have frequently been imposed (Bai and Ng 2002; Johnson and Wichern 2007). Condition C3 was imposed in Fan, Liao, and Mincheva (2013); it ensures that factor model (1) is asymptotically identifiable and implies that $\epsilon_1, \ldots, \epsilon_p$ are weakly correlated after removing the common factors. They are related to identifiability and moment conditions. Condition C4 shows that $\mathbf{B}$ is of full column rank, that is, $\mathrm{rank}(\mathbf{B}) = K$. Condition C5 shows that every factor f_j has an impact on at least two observed variables.

By (6), the population correlation matrix of $\mathbf{y}$ in the factor model (1) is

$$\mathbf{R} = \mathbf{Q}\mathbf{Q}^T = \mathbf{Q}_1\mathbf{Q}_1^T + \mathbf{Q}_2\mathbf{Q}_2^T, \tag{9}$$

where $\mathbf{Q} = [\mathrm{diag}(\boldsymbol{\Sigma})]^{-1/2}(\mathbf{B}, \boldsymbol{\Psi}^{1/2}) = (\mathbf{Q}_1, \mathbf{Q}_2)$ with

$$\mathbf{Q}_1 = [\mathrm{diag}(\boldsymbol{\Sigma})]^{-1/2}\mathbf{B}, \quad \mathbf{Q}_2 = [\mathrm{diag}(\boldsymbol{\Sigma})]^{-1/2}\boldsymbol{\Psi}^{1/2}. \tag{10}$$

Let $\|\mathbf{M}\|_F$ denote the Frobenius norm of a matrix $\mathbf{M}$ and $\|\mathbf{M}\| = \sqrt{\lambda_1(\mathbf{M}\mathbf{M}^T)}$ the operator norm. The following theorem shows how to determine the number of factors from the population correlation matrix.

Theorem 1. Under Conditions C1–C5, if $\|[\mathrm{diag}(\boldsymbol{\Sigma})]^{-1}\boldsymbol{\Psi}\| \le 1$, we have

$$\lambda_j(\mathbf{R}) \le 1, \quad j = K+1, \ldots, p.$$

In addition, we have

$$K = \max\{j : \lambda_j(\mathbf{R}) > 1, j \in [p]\}, \tag{11}$$

when there exist three nonnegative constants $\delta_1 > \delta_2 + \delta_3 \ge 0$, $\delta_3 < 0.5$ satisfying as $p \to \infty$

$$\begin{aligned} &\|[\mathrm{diag}(\boldsymbol{\Sigma})]^{-1/2}\mathbf{B}\|_F^2 = O(p^{\delta_1}), \quad K = O(p^{\delta_3}), \\ &\|\mathbf{B}^T[\mathrm{diag}(\boldsymbol{\Sigma})]^{-1}\mathbf{B}\| \cdot \|\{\mathbf{B}^T[\mathrm{diag}(\boldsymbol{\Sigma})]^{-1}\mathbf{B}\}^{-1}\| = O(p^{\delta_2}), \\ &\|[\mathrm{diag}(\boldsymbol{\Sigma})]^{-1}\boldsymbol{\Psi}\| \le 1. \end{aligned} \tag{12}$$

Remark 1. To understand (12), without loss of generality, assume $\mathrm{diag}(\boldsymbol{\Sigma}) = \mathbf{I}_p$. Then $\mathbf{R} = \mathbf{B}\mathbf{B}^T + \boldsymbol{\Psi}$ and the conditions (12) become

$$\begin{aligned} &K = O(p^{\delta_3}), \ \|\boldsymbol{\Psi}\| \le 1, \ \|\mathbf{B}\|_F^2 = O(p^{\delta_1}), \\ &\|\mathbf{B}^T\mathbf{B}\| \cdot \|(\mathbf{B}^T\mathbf{B})^{-1}\| = O(p^{\delta_2}). \end{aligned} \tag{13}$$

In fact, when $\mathbf{\Psi}$ is a diagonal matrix, $\|\mathbf{\Psi}\| \leq 1$ holds automatically. If the factor loadings $\{\tilde{\mathbf{b}}_i\}_{i=1}^p$ ($\tilde{\mathbf{b}}_i^T$ is the ith row of $\mathbf{B}$) are a random sample from a population with $\mathrm{E}(\tilde{\mathbf{b}}_i\tilde{\mathbf{b}}_i^T) = \mathbf{\Sigma}_b$, when K is fixed, by the strong law of large numbers, we have

$$p^{-1}\mathbf{B}^T\mathbf{B} = p^{-1}\sum_{i=1}^{p}\tilde{\mathbf{b}}_i\tilde{\mathbf{b}}_i^T \to \mathbf{\Sigma}_b, \quad \text{almost surely.}$$

Hence, condition (13) holds for almost all such realizations with $\delta_1 = 1$, $\delta_2 = 0$, and $\delta_3 = 0$. Indeed, the above arguments are valid so long as $\delta_3 < 1$ and additional conditions on the distribution of $\tilde{\mathbf{b}}_i$ (e.g. sub-Gaussian) are imposed.

3. Properties of Sample Correlation Matrix Under Factor Model

Let $\mathbf{y}_1, \ldots, \mathbf{y}_n$ be an iid sample of size n from (1):

$$\mathbf{y}_i = \boldsymbol{\alpha} + \mathbf{B}\mathbf{f}_i + \boldsymbol{\epsilon}_i,\ i \in [n] = \{1, \ldots, n\}.$$

Then the sample covariance matrix and sample correlation matrix are

$$\begin{aligned} \hat{\mathbf{\Sigma}}_n &= n^{-1}\sum_{i=1}^{n}(\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^T, \\ \hat{\mathbf{R}} &= [\mathrm{diag}(\hat{\mathbf{\Sigma}}_n)]^{-1/2}\hat{\mathbf{\Sigma}}_n[\mathrm{diag}(\hat{\mathbf{\Sigma}}_n)]^{-1/2}, \end{aligned} \tag{14}$$

where $\bar{\mathbf{y}} = n^{-1}\sum_{i=1}^n \mathbf{y}_i$ is the sample mean. Let the empirical spectral distributions (ESDs) of $\hat{\mathbf{R}}$ and $\mathbf{R}$ be $F_n(t)$ and $H_{p-K}(t)$ as follows:

$$F_n(t) = \frac{1}{p-K}\sum_{j=K+1}^{p} 1(\lambda_j(\hat{\mathbf{R}}) \leq t), \tag{15}$$

$$H_{p-K}(t) = \frac{1}{p-K}\sum_{j=K+1}^{p} 1(\lambda_j(\mathbf{R}) \leq t),$$

for any real number t, where $1(\cdot)$ is an indicator function.

3.1. Spectral Properties of Sample Correlation Matrix

To estimate the number of common factors, we first derive some fundamental results in random matrix theory: the Stieltjes equation of the limiting spectral distribution (LSD) $F(t)$ of the ESD $F_n(t)$, and the almost sure convergence of sample spiked eigenvalues $\lambda_1(\hat{\mathbf{R}}), \ldots, \lambda_K(\hat{\mathbf{R}})$ of $\hat{\mathbf{R}}$. There are some reports in the literature on the spectral properties of $\hat{\mathbf{R}}$ when $\mathbf{R}$ has special structures. Bao, Pan, and Zhou (2012) derived the Tracy–Widom law of the maximum eigenvalue of $\hat{\mathbf{R}}$ when $\mathbf{R} = \mathbf{I}_p$ and $p/n \to \rho \in (0, \infty)$. El Karoui (2007) established the LSD of $\hat{\mathbf{R}}$ for the elliptical distribution as $p/n \to \rho \in (0, \infty)$ with the bounded spectral norm $\|\mathbf{R}\|$. Gao et al. (2017) obtained the central limit theorem of $\hat{\mathbf{R}}$ for the case $\mathbf{R} = \mathbf{I}_p$ and $p/n \to \rho \in (0, \infty)$. However, for the general factor model (1), the population correlation matrix is not $\mathbf{I}_p$. Theorem 2 gives the Stieltjes equation of the LSD of $\hat{\mathbf{R}}$ in the general case. For convergence of sample spiked eigenvalues $\lambda_1(\hat{\mathbf{R}}), \ldots, \lambda_K(\hat{\mathbf{R}})$, we have not found the related literature; Theorem 3 fills the void.

To derive Theorems 2 and 3, additional assumptions are needed.

Assumption (a). Letting $\mathbf{x}_i = (x_{1i}, \ldots, x_{p+K,i})^T = (f_{1i}, \ldots, f_{Ki}, e_{1i}, \ldots, e_{pi})^T$ and $(e_{1i}, \ldots, e_{pi}) = (\epsilon_{1i}, \ldots, \epsilon_{pi})\mathbf{\Psi}^{-1/2}$, $\{x_{ji}, j \in [p+K], i \in [n]\}$ are independent random variables satisfying

$$\frac{1}{n(p+K)\eta_n^4}\sum_{j=1}^{p+K}\sum_{i=1}^{n}\mathrm{E}|x_{ji}^4|1(|x_{ji}| > \eta_n\sqrt{n}) \to 0, \tag{16}$$

where $\{\eta_n\}$ is a given sequence of positive numbers satisfying $\eta_n \to 0$ and $\eta_n \log n \to +\infty$.

Assumption (b). $\sup_{j\in[p+K]} \mathrm{E}(|x_{j1}|^{6+\delta_0})$ is bounded for all p for some $\delta_0 > 0$.

Assumption (c). $\rho_n = p/n \to \rho \in (0, \infty)$ as $n \to \infty$.

Assumption (d). The number K of common factors is fixed.

Assumption (e). $\|[\mathrm{diag}(\mathbf{\Sigma})]^{-1}\mathbf{\Psi}\| \leq 1$ and the LSD $H(t)$ of the ESD $H_{p-K}(t)$ from the eigenvalues $\lambda_{K+1}(\mathbf{R}), \ldots, \lambda_p(\mathbf{R})$ of $\mathbf{R}$ exists.

Remark 2. The assumption (16) is the Lindeberg condition. As we impose weak moment conditions, we need to use the truncation tricks and hence the proofs of Theorems 2 and 3 are somewhat lengthy and can be found in the supplementary materials. By Theorem 1, $\lambda_j(\mathbf{R}) \leq 1$ for $j = K+1, \ldots, p$ if $\|[\mathrm{diag}(\mathbf{\Sigma})]^{-1}\mathbf{\Psi}\| \leq 1$. Thus, the support set of $H(t)$ is in $[0, 1]$.

To derive the convergence of sample spiked eigenvalues $\lambda_j(\hat{\mathbf{R}}), j \in [K]$ and obtain the consistent estimators of $\lambda_j, j \in [K]$, we need the Stieltjes transform of the ESD of the sample correlation matrix $\hat{\mathbf{R}}$. For $z \in \mathcal{C}^+$, the Stieltjes transforms $m_n(z)$, $\underline{m}(z)$, $m(z)$, and $\underline{m}(z)$ are defined as follows:

$$\begin{aligned} m_n(z) &= (p-K)^{-1}\sum_{j=K+1}^{p}(\lambda_j(\hat{\mathbf{R}}) - z)^{-1} = \int \frac{1}{t-z}dF_n(t), \\ \underline{m}_n(z) &= \int \frac{1}{t-z}d\underline{F}_n(t) = -(1-\rho_{K,n-1})z^{-1} + \rho_{K,n-1}m_n(z), \\ m(z) &= \int \frac{1}{t-z}dF(t), \\ \underline{m}(z) &= \int \frac{1}{t-z}d\underline{F}(t) = -(1-\rho)z^{-1} + \rho m(z), \end{aligned} \tag{17}$$

where $F_n(z)$ is defined in (15), $\rho_{K,n-1} = (p-K)/(n-1)$, $\underline{F}_n(x) = (1-\rho_{K,n-1})1(x > 0) + \rho_{K,n-1}F_n(x)$, $\underline{F}(x) = (1-\rho)1(x > 0) + \rho F(x)$ and $\mathcal{C}^+$ denotes the upper plane of the two-dimensional complex space. Then $\underline{m}(z)$ and $m(z)$ satisfy the equations (18) and (19). Define the notation $o_{a.s.}(1)$ as the random variable going to zero almost surely and define $o_p(1)$ as the random variable going to zero in probability.

Theorem 2. For the high-dimensional factor model (1) satisfying Conditions C1–C5 and Assumptions (a)–(e), we have

$$|m_n(z) - m(z)| = o_{\mathrm{a.s.}}(1), \quad |\underline{m}_n(z) - \underline{m}(z)| = o_{\mathrm{a.s.}}(1),$$

$$z = -\frac{1}{\underline{m}(z)} + \rho\int\frac{tdH(t)}{1+t\underline{m}(z)} \text{ or } \frac{z}{-\underline{m}^{-1}(z)} = \psi(-\underline{m}^{-1}(z)), \tag{18}$$

where $z \in \mathcal{C}^+$ and

$$\psi(x) = 1 + \rho\int\frac{t}{x-t}dH(t). \tag{19}$$

Remark 3. Theorem 2 proves that the Stieltjes transforms $m_n(z)$ and $\underline{m}_n(z)$ of the ESDs $F_n(x)$ and $\underline{F}_n(x)$ converge to the Stieltjes transforms $m(z)$ and $\underline{m}(z)$ of the LSDs $F(x)$ and $\underline{F}(x)$ almost surely. Moreover, (18) gives the explicit expression of the Stieltjes transform $\underline{m}(z)$ of the LSD $\underline{F}(x)$. It indicates the convergence of sample spiked eigenvalues $\hat{\lambda}_j = \lambda_j(\hat{\mathbf{R}}), j \in [K]$ and motivates a consistent estimator of $\lambda_j = \lambda_j(\mathbf{R}), j \in [K]$. Theorem 3 will show that $-\underline{m}_{n,j}^{-1}(z)/\lambda_j = 1 + o_p(1)$ and $\hat{\lambda}_j/\lambda_j = \psi(\lambda_j) + o_p(1)$ for $j \in [K]$, where $\underline{m}_{n,j}$ is an estimator of $\underline{m}(z)$.

3.2. Bias Correction of Sample Eigenvalues

Let $\hat{\lambda}_j = \lambda_j(\hat{\mathbf{R}})$ and $\lambda_j = \lambda_j(\mathbf{R})$ for $j \in [p]$. For any given j, define

$$m_{n,j}(z) = (p-j)^{-1}\Bigg[\sum_{\ell=j+1}^{p}(\hat{\lambda}_\ell - z)^{-1} + ((3\hat{\lambda}_j + \hat{\lambda}_{j+1})/4 - z)^{-1}\Bigg],$$

$$\underline{m}_{n,j}(z) = -(1-\rho_{j,n-1})z^{-1} + \rho_{j,n-1}m_{n,j}(z),$$

with $\rho_{j,n-1} = (p-j)/(n-1)$. Let the corrected eigenvalue of $\hat{\lambda}_j$ be

$$\hat{\lambda}_j^C = -\frac{1}{\underline{m}_{n,j}(\hat{\lambda}_j)}.$$

This estimator was introduced in Bai and Ding (2012). Consistency is established when $\mathbf{\Sigma}$ is block diagonal and λ_j is bounded. The following theorem, whose proof is given in the supplementary materials, shows that the corrected eigenvalues are consistent.

Theorem 3. For the high-dimensional factor model (1) satisfying Conditions C1–C5 and Assumptions (a)–(e), for $j \in [K]$, if $\lambda_j \geq \lambda_{K+1}(\mathbf{R})(1+\sqrt{\rho}) + \delta$ for some $\delta > 0$,

$$\frac{\hat{\lambda}_j^C}{\lambda_j} = 1 + o_p(1) \quad \text{and} \quad \frac{\hat{\lambda}_j}{\lambda_j} = \psi(\lambda_j) + o_p(1). \tag{20}$$

In particular, if in addition λ_j is bounded for $j \in [K]$, we have

$$\hat{\lambda}_j^C = \lambda_j + o_p(1) \quad \text{and} \quad \hat{\lambda}_j = \lambda_j\psi(\lambda_j) + o_p(1).$$

Remark 4. By Remark 2 and (11), we have $\lambda_j > 1$ for $j \in [K]$ under the conditions of Theorem 1 and the support of $H(t)$ being in $[0,1]$. By (19) and (20), if $\lambda_j > \lambda_{K+1}(\mathbf{R})(1+\sqrt{\rho}) + \delta$ for some $\delta > 0$ and is bounded, we have

$$\hat{\lambda}_j - \lambda_j = \lambda_j\psi(\lambda_j) - \lambda_j = \rho\int\frac{\lambda_j t}{\lambda_j - t}dH(t) + o_p(1),\ j \in [K].$$

In other words, the sample eigenvalue $\hat{\lambda}_j$ is not a consistent estimator of λ_j for $j \in [K]$. This is due to the inconsistency of the high-dimensional sample correlation matrix. On the other hand, from (20), we show that the corrected eigenvalue $\hat{\lambda}_j^C$ is consistent for $j \in [K]$.

4. Minimum Signals and Optimal Threshold

This section describes how to find minimal signal strength ν_0 for consistent estimation of the number of common factors and to give the optimal threshold level for our estimator.

4.1. Minimal Signal Strength

The following theorem shows the minimal signal strength.

Theorem 4 (Minimal signal strength ν_0). For a high-dimensional factor model (1) satisfying Conditions C1–C5 and Assumptions (a)–(e), and for any estimation method $\hat{K}_{\text{any}}$ of the number of common factors by detecting the gap between $\{\lambda_j(\hat{\mathbf{R}}), j \in [K]\}$ and $\{\lambda_j(\hat{\mathbf{R}}), j = K+1, \ldots, p\}$ (i.e., $\lambda_K(\hat{\mathbf{R}}) - \lambda_{K+1}(\hat{\mathbf{R}}) \neq o_p(1)$), if $\nu_0 < 1 + \sqrt{\rho}$, then

$$\limsup_{n\to\infty}\inf_{\mathbf{R}\in\mathcal{F}(\nu_0)} P(\hat{K}_{\text{any}} = K) < 1.$$

Proof. Let us take ϵ such that $(1-\epsilon)(1+\sqrt{\rho}) > \max\{1, \nu_0\}$ and $\mathbf{R} \in \mathcal{F}(\nu_0)$ such that $\mathbf{R} = \mathbf{\Sigma} = \text{diag}(\mathbf{R})$ and

$$\lambda_p(\mathbf{R}) = \cdots = \lambda_{K+1}(\mathbf{R}) = 1-\epsilon < \lambda_K(\mathbf{R}) = (1-\epsilon)(1+\sqrt{\rho}).$$

Then by (20), we have $\lambda_K(\hat{\mathbf{R}}) = (1-\epsilon)(1+\sqrt{\rho})^2 + o_p(1)$ because $H(t)$ is the limit of the empirical distribution function of $\{1-\epsilon, \ldots, 1-\epsilon\}$. By (S.46) in the supplementary materials, we have $|\lambda_{K+1}(\hat{\mathbf{R}}) - \lambda_{K+1}(\mathbf{S}_n)| = o_{\text{a.s.}}(1)$. By Theorem 1.1 of Baik and Silverstein (2006), we have $(1-\epsilon)^{-1}\lambda_{K+1}(\mathbf{S}_n) = (1+\sqrt{\rho})^2 + o_{\text{a.s.}}(1)$. Thus, we have

$$\lambda_{K+1}(\hat{\mathbf{R}}) = (1-\epsilon)(1+\sqrt{\rho})^2 + o_{\text{a.s.}}(1). \tag{21}$$

Hence, when n, p are large enough, $\lambda_K(\hat{\mathbf{R}})$ and $\lambda_{K+1}(\hat{\mathbf{R}})$ will be indifferentiable. That is, when n, p are large enough, the difference between $\lambda_K(\hat{\mathbf{R}})$ and $\lambda_{K+1}(\hat{\mathbf{R}})$ cannot be detected by any method. □

Remark 5. The above theorem shows that $\nu_0 = 1 + \sqrt{\rho}$ is the minimal signal strength. Thus, throughout the rest of the article, we will consider the following estimation method in the set of the correlation matrix $\mathbf{R}$ as follows:

$$\mathcal{F}(1+\sqrt{\rho}) = \{\mathbf{R}:\ \mathbf{R} \text{ is the correlation matrix (9) of the observed vector in the factor model (1) and } \lambda_K(\mathbf{R}) > 1 + \sqrt{\rho}\}.$$

4.2. Optimal Threshold

Recall our estimation method,

$$\hat{K}^C(s) = \max\{j : \hat{\lambda}_j^C > s\}, \tag{22}$$

where the maximum of the empty set is defined as 0. The following theorem establishes the optimal bound of the threshold s.

Theorem 5. For the high-dimensional factor model (1) satisfying Conditions C1–C5 and Assumptions (a)–(e), we have

$$\limsup_{n\to\infty}\sup_{\mathbf{R}\in\mathcal{F}(1+\sqrt{\rho})} P(\hat{K}^C(s) > K) = 1, \quad \text{if } s < 1+\sqrt{\rho}, \tag{23}$$

$$\limsup_{n\to\infty}\sup_{\mathbf{R}\in\mathcal{F}(1+\sqrt{\rho})} P(\hat{K}^C(s) < K) = 1, \quad \text{if } s > 1+\sqrt{\rho}, \tag{24}$$

where s does not depend on n and p.

Proof. To (23), let $(1-\epsilon)(1+\sqrt{\rho}) > s$ and $\mathbf{R} \in \mathcal{F}(1+\sqrt{\rho})$ satisfy

$$\lambda_p(\mathbf{R}) = \cdots = \lambda_{K+1}(\mathbf{R}) = 1-\epsilon < 1+\sqrt{\rho} < \lambda_K(\mathbf{R}).$$

By (18) and (20), we have $\lambda_{K+1}(\hat{\mathbf{R}}) = \hat{\lambda}^C_{K+1} + \rho\frac{\hat{\lambda}^C_{K+1}(1-\epsilon)}{\hat{\lambda}^C_{K+1}-(1-\epsilon)} + o_p(1)$, because $H(t)$ is the limit of the empirical distribution function of $\{1-\epsilon,\ldots,1-\epsilon\}$. It then follows from $\lambda_{K+1}(\hat{\mathbf{R}}) = (1-\epsilon)(1+\sqrt{\rho})^2 + o_p(1)$ that

$$\hat{\lambda}^C_{K+1} = (1-\epsilon)(1+\sqrt{\rho}) + o_p(1).$$

That is, $\hat{\lambda}^C_j \geq (1-\epsilon)(1+\sqrt{\rho}) + o_p(1)$, $j \in [K+1]$. By using $(1-\epsilon)(1+\sqrt{\rho}) > s$, we conclude that

$$\limsup_{n\to\infty} \sup_{\mathbf{R}\in\mathcal{F}(1+\sqrt{\rho})} P(\hat{K}^C(s) > K) = 1,$$

when $s < 1+\sqrt{\rho}$.

To prove (24), let $\mathbf{\Sigma} = \mathbf{R} \in \mathcal{F}(1+\sqrt{\rho})$ satisfy

$$\lambda_p(\mathbf{R}) \leq \cdots \leq \lambda_{K+1}(\mathbf{R}) \leq 1 < 1+\sqrt{\rho} < \lambda_K(\mathbf{R}) < s \leq \cdots \leq \lambda_1(\mathbf{R}).$$

By (20) and $\lambda_K(\mathbf{R}) < s$, we have $\hat{\lambda}^C_K = \lambda_K(\mathbf{R}) + o_p(1) < s + o_p(1)$ which means $\hat{\lambda}^C_j < s$, $j = p,\ldots,K$ in probability. Thus,

$$\limsup_{n\to\infty} \sup_{\mathbf{R}\in\mathcal{F}(1+\sqrt{\rho})} P(\hat{K}^C(s) < K) = 1,$$

if $s > 1+\sqrt{\rho}$. □

Theorem 5 shows that the choices $s < 1+\sqrt{\rho}$ and $s > 1+\sqrt{\rho}$ are not optimal for the threshold parameter s in our estimation method. The following theorem will show that $s = 1+\sqrt{\rho}$ is optimal.

Theorem 6. For the high-dimensional factor model (1) satisfying Conditions C1–C5 and Assumptions (a)–(e), for $\mathbf{R} \in \mathcal{F}(1+\sqrt{\rho})$, we have when $s = 1+\sqrt{\rho}+\delta < \lambda_K(\mathbf{R})$ for a very small positive number δ,

$$P(\hat{K}^C(s) = K) \to 1. \tag{25}$$

Proof. For any $\mathbf{R} \in \mathcal{F}(1+\sqrt{\rho})$, we have

$$\lambda_p(\mathbf{R}) \leq \cdots \leq \lambda_{K+1}(\mathbf{R}) \leq 1 < 1+\sqrt{\rho}+\epsilon_0 < \lambda_K(\mathbf{R}) \leq \cdots \leq \lambda_1(\mathbf{R}),$$

for a very small positive number $\epsilon_0 > \delta$. By (18), we have $\lambda_{K+1}(\hat{\mathbf{R}}) = \hat{\lambda}^C_{K+1} + \rho\int\frac{\hat{\lambda}^C_{K+1}t}{\hat{\lambda}^C_{K+1}-t}dH(t) + o_p(1)$. Thus, we have

$$\hat{\lambda}^C_{K+1} \leq 1+\sqrt{\rho}+o_p(1), \tag{26}$$

because of $\lambda_{K+1}(\hat{\mathbf{R}}) \leq (1+\sqrt{\rho})\psi(1+\sqrt{\rho}) + o_p(1)$ by Lemma S.6 in the supplementary materials. By (20), we have

$$\hat{\lambda}^C_j \geq 1+\sqrt{\rho}+\epsilon_0+o_p(1),\ j \in [K]. \tag{27}$$

Thus, by (26) and (27), when $s = 1+\sqrt{\rho}+\delta$, we have

$$\lim_{n\to\infty} P(\hat{K}^C(s) = K) = 1.$$

□

5. Simulation Studies

We evaluate the finite-sample performance of the proposed method by simulation studies. As our proposed method is based on adjusted correlation thresholding, we denote the proposed estimation method "ACT." We compare ACT with 13 other methods: "PC_1," "PC_2,""PC_3,""IC_1," "IC_2" and "IC_3" from Bai and Ng (2002), "ON_1," "ON_2," and "ON_3" from Onatski (2005), "NON" from Onatski (2010), "ER" and "GR" from Ahn and Horenstein (2013). The simulation results indicate that PC_3 performs the best among PC_1, PC_2, PC_3, IC_1, IC_2, IC_3. Hence, of the six methods of Bai and Ng (2002), we only show the results for PC_3. Owing to the similarity of simulation results, we only present PC_3, ON_2, ER, GR and ACT.

The sample sizes are taken to be $n = 150, 300$ and the dimensions are $p = 100, 300, 500, 1000$. We set the true number of common factors to be $K = 5$. For the Gaussian population, we draw ϵ_i's iid from $N(0, \nu_i^2)$ and $f_1,\ldots,f_K$ iid from $N(0,1)$. For every case, we conduct 1000 replications and summarize the empirical percentages of true estimation, overestimation, underestimation of the number of common factors, and the average number of common factors. We consider the following five cases for the factor loading matrix $\mathbf{B} = (b_{\ell j})_{\ell\in[p], j\in[K]}$. They can be verified to satisfy the imposed conditions, in particular, those in Theorem 1.

Case 1: Let $b_{\ell j} = \sqrt{3p^{-1/2}}$ for $\ell, j \in [K]$ and $b_{\ell j} = a_{\ell j}\sqrt{3(p-j)^{-1}}$ for $\ell \in \{K+1,\ldots,p\}, j \in [K]$ and $a_{\ell j} = -1$ if $\ell = rj$ or $a_{\ell j} = 1$ if $\ell \neq rj$ for $r = 1,2,3\ldots$. Assume that $\nu_1^2 = \cdots = \nu_p^2 = 0.55^2$. The model is from Harding (2013).

Case 2: Let $b_{\ell j}$ be iid from $N(0,1)$ for $\ell \in [p], j \in [K]$ and $\nu_1^2 = \cdots = \nu_p^2 = 180$. The model is utilized in Bai and Ng (2002) and Onatski (2010).

Case 3: Let $b_{jj} = 1$, $b_{\ell j}$ be iid from $N(0, 0.04)$ for $j \neq \ell$, $\ell \in [p], j \in [K]$ and $\nu_1^2,\ldots,\nu_p^2$ be iid from Unif(0, 5.5).

Case 4: Let $b_{\ell j}$ be iid from $N(0,1)$ for $j \in [K]$, $\ell \in [\rho p]$ and $b_{\ell j}$ be iid from Unif(0, 0.05) for $j \in [K]$, $\ell \in \{\rho p+1,\ldots,p\}$. Let $\nu_1^2,\ldots,\nu_p^2$ be iid from Unif(0, 20). The parameter ρ is taken as $\rho = 0.15, 0.3$ or 0.5.

Case 5: For $1 \leq j \leq 2$, let $b_{\ell j}$ be iid from $N(0, 25)$ for $\ell \in [\rho p]$ and $b_{\ell j}$ be iid from Unif(0, 0.05) for $\ell \in \{\rho p+1,\ldots,p\}$. For $3 \leq j \leq 5$, let $b_{\ell j}$ be iid from $N(0,1)$ for $\ell \in [p]$. Let $\nu_1^2,\ldots,\nu_p^2$ be iid from Unif(0, 180). The parameter ρ is taken as $0.05, 0.15$ or 0.5.

Case 1 has all weak factors, Cases 2 and 3 have all pervasive factors, and Cases 4 and 5 have a mix of weak and strong factors. For Cases 3 and 4, the variances $\nu_1^2,\ldots,\nu_p^2$ of the error variables $\epsilon_1,\ldots,\epsilon_p$ are different. The simulation results for Cases 1–5 with $n = 300$ are presented in Tables 2–6, respectively. The results for the cases with $n = 150$ are similar and are omitted. For the uniform population that ϵ_i are iid from Unif$(-\sqrt{3\nu_i^2}, \sqrt{3\nu_i^2})$ and $f_1,\ldots,f_K$ are iid from Unif$(-\sqrt{3}, \sqrt{3})$, the results can be found in Section S.7 of the supplementary materials. As shown in these tables, except for a very few settings, "ACT" outperforms competing methods. Even for these few settings, the percentiles of true estimation of "ACT" are also similar to those of "ER" and "GR."

Table 2. Percentages of the estimated number of common factors for Case 1 with $n = 300$ in 1000 simulations: "$\hat{K} = K$," "$\hat{K} > K$," and "$\hat{K} < K$" truly estimates, overestimates, and underestimates the number of common factors, respectively.

p		PC_3	ON_2	ER	GR	ACT
		Gaussian population				
100	$\hat{K} = K$	99.8	100	41.8	75.7	100
	$\hat{K} > K$	0	0	0	0	0
	$\hat{K} < K$	0.2	0	58.2	24.3	0
	ave($\hat{K}$)	5	5	2.87	4.37	5
300	$\hat{K} = K$	92.0	99.9	4.2	8.7	100
	$\hat{K} > K$	0	0.1	0	0	0
	$\hat{K} < K$	8.0	0	95.8	91.3	0
	ave($\hat{K}$)	4.92	5	2.09	2.56	5
500	$\hat{K} = K$	0	100	0	0.2	99.6
	$\hat{K} > K$	0	0	0	0	0.4
	$\hat{K} < K$	100	0	100	99.8	0
	ave($\hat{K}$)	3.88	5	1.78	1.96	5
1000	$\hat{K} = K$	0	79.1	0	0	89.0
	$\hat{K} > K$	0	0	0	0	1.9
	$\hat{K} < K$	100	20.9	100	100	9.1
	ave($\hat{K}$)	1.81	4.79	1.34	1.38	4.93

NOTE: "ave($\hat{K}$)" is the average of the estimated number of common factors.

Table 3. Percentages of the estimated number of common factors for Cases 2 and 3 with $n = 300$ in 1000 simulations: "$\hat{K} = K$," "$\hat{K} > K$," and "$\hat{K} < K$" truly estimates, overestimates, and underestimates the number of common factors, respectively.

p		PC_3	ON_2	ER	GR	ACT
		Gaussian population for Case 2				
100	$\hat{K} = K$	0	0.1	5.5	5.8	0
	$\hat{K} > K$	0	0	9.6	9.7	0
	$\hat{K} < K$	100	99.9	84.9	84.5	100
	ave($\hat{K}$)	1	1.27	2.51	2.54	1.06
300	$\hat{K} = K$	0	1.1	4.2	4.6	5.4
	$\hat{K} > K$	0	0	0.8	0.9	0
	$\hat{K} < K$	100	98.9	95	94.5	94.6
	ave($\hat{K}$)	1	2.85	2.1	2.14	2.91
500	$\hat{K} = K$	0	32.5	26.0	27.3	71.3
	$\hat{K} > K$	0	0	0.2	0.2	2.8
	$\hat{K} < K$	100	67.5	73.8	72.5	25.9
	ave($\hat{K}$)	1	4.2	2.92	2.97	4.74
1000	$\hat{K} = K$	0	99.6	92.3	92.7	96.2
	$\hat{K} > K$	0	0	0	0	3.8
	$\hat{K} < K$	100	0.4	7.7	7.3	0
	ave($\hat{K}$)	1	5	4.81	4.83	5.04
		Gaussian population for Case 3				
100	$\hat{K} = K$	0.2	0.7	3.9	4.6	98.2
	$\hat{K} > K$	0	0	1.9	2.4	0.2
	$\hat{K} < K$	99.8	99.3	94.2	93	1.6
	ave($\hat{K}$)	2.4	2.85	2.14	2.21	4.99
300	$\hat{K} = K$	99.5	97.8	81.6	83	99.3
	$\hat{K} > K$	0.1	0.1	0	0	0.7
	$\hat{K} < K$	0.4	2.1	18.4	17.0	0
	ave($\hat{K}$)	5	4.98	4.55	4.6	5.01
500	$\hat{K} = K$	63.9	100	99.9	99.9	99.4
	$\hat{K} > K$	0	0	0	0	0.6
	$\hat{K} < K$	36.1	0	0.1	0.1	0
	ave($\hat{K}$)	4.63	5	5	5	5.01
1000	$\hat{K} = K$	4.9	99.9	100	100	99.5
	$\hat{K} > K$	0	0.1	0	0	0.5
	$\hat{K} < K$	95.1	0.0	0	0	0
	ave($\hat{K}$)	3.6	5	5	5	5

NOTE: "ave($\hat{K}$)" is the average of the estimated number of common factors.

6. Empirical Studies

This section analyzes two real datasets on economics and finance to demonstrate our proposed estimation method ACT.

Example 1 (Macroeconomic time series). We use the monthly macroeconomic datasets from March 1960 to December 2014 utilized by McCraken and Ng (2017). Series 64, 66, 101, and 130 are removed because of missing observations. Following McCraken and Ng (2017), outliers are removed where an outlier is defined as an observation that deviates from the sample mean by more than ten interquartile ranges. After the datasets are cleaned, the data dimension is $p = 123$ and the sample size is $n = 583$. McCraken and Ng (2017) used PC_2 to select nine factors by using the sample covariance matrix whose nine largest eigenvalues are 3.91×10^{11}, 1.20×10^{10}, 4.77×10^{9}, 3.06×10^{9}, 7.25×10^{8}, 3.60×10^{8}, 1.38×10^{8}, 2.83×10^{7}, and 9.00×10^{6}. However, the marginal variances of these 123 time series vary widely from 2.80×10^{-4} to 1.80×10^{11}, which jeopardize the fidelity of the covariance matrix based methods. Our estimation method ACT selects six factors by using the sample correlation matrix whose top nine eigenvalues are 73.10, 17.81, 10.22, 7.00, 4.80, 1.97, 1.53, 1.17, 1.06.

In terms of percentage of variance explained by the selected factors, the nine selected factors explain 99.99% of total variation, whereas the six selected factors explain 99.95% of total variation. This is mainly due to the leading eigenvalue which is an order of magnitude larger than the rest. If we look at the standardized variables (the eigenvalues from the correlation matrix), the selected nine factors explain 96.49% of total variations whereas the selected six factors explain 93.43%.

We now examine whether the number of factors that influences the equity market has changed before and after financial crisis. As an illustration, we use the stationarily transformed macroeconomic time series (McCraken and Ng 2017).

6.1. Transformed Macro Data Before the Financial Crisis

We now use the stationarily transformed (McCraken and Ng 2017) monthly macroeconomic datasets from January 1960 to December 2007 with sample size $n = 576$ and $p = 123$. Using PC_2 as in McCraken and Ng (2017), nine factors are selected. The nine largest eigenvalues for the sample covariance are 2.28×10^{4}, 13.06, 1.53, 0.88, 0.74, 0.40, 0.32, 0.24, 0.18. Again, the marginal sample variances for these 123 transformed series vary widely from 8.47×10^{-7} to 2.28×10^{4}. On the other hand, our estimation method ACT selects 10 factors by using the sample correlation matrix. The 10 largest eigenvalues of the sample correlation matrix are 18.37, 8.55, 7.66, 6.16, 5.86, 4.01, 3.76, 3.53, 2.89, 2.56. The variances explained by nine selected and ten selected factors are both around 99.99% due to a very spike top eigenvalue. In terms of percentage of variance explained by the standardized variables, ten factors explain 51.53% whereas nine factors explain 49.45%.

Table 4. Percentages of the estimated number of common factors for Case 4 with $n = 300$ in 1000 simulations: "$\hat{K} = K$," "$\hat{K} > K$," and "$\hat{K} < K$" truly estimates, overestimates, and underestimates the number of common factors, respectively.

			PC_3	ON_2	ER	GR	ACT
ρ	p		Gaussian population				
$\rho = 0.15$	100	$\hat{K} = K$	0.1	0	0.6	0.6	0
		$\hat{K} > K$	0	0	0.1	0.1	0
		$\hat{K} < K$	99.9	100	99.3	99.3	100
		ave($\hat{K}$)	1.84	2.28	1.41	1.43	1.97
	300	$\hat{K} = K$	50.1	35.8	5.4	5.9	64.4
		$\hat{K} > K$	0	0	0	0	1.3
		$\hat{K} < K$	49.9	64.2	94.6	94.1	34.3
		ave($\hat{K}$)	4.47	4.29	2.22	2.29	4.66
	500	$\hat{K} = K$	7.7	98.4	60.0	63.2	96.6
		$\hat{K} > K$	0	0.2	0	0	3.2
		$\hat{K} < K$	92.3	1.4	40.0	36.8	0.2
		ave($\hat{K}$)	3.77	4.99	4.02	4.13	5.03
$\rho = 0.3$	100	$\hat{K} = K$	7.6	12.0	1.4	1.6	32.9
		$\hat{K} > K$	0	0	0	0	0.1
		$\hat{K} < K$	92.4	88.0	98.6	98.4	67.0
		ave($\hat{K}$)	3.66	3.84	2.07	2.15	4.22
	300	$\hat{K} = K$	99.7	99.9	95.8	96.3	99.0
		$\hat{K} > K$	0.3	0.1	0	0	1.0
		$\hat{K} < K$	0	0	4.2	3.7	0
		ave($\hat{K}$)	5	5	4.92	4.94	5.01
	500	$\hat{K} = K$	100	99.9	99.9	99.9	98.9
		$\hat{K} > K$	0	0.1	0	0	1.1
		$\hat{K} < K$	0	0	0.1	0.1	0
		ave($\hat{K}$)	5	5	5	5	5.01
$\rho = 0.5$	100	$\hat{K} = K$	90.4	93.1	53.9	57.5	99.4
		$\hat{K} > K$	0.1	0	0	0	0.4
		$\hat{K} < K$	9.5	6.9	46.1	42.5	0.2
		ave($\hat{K}$)	4.91	4.93	3.93	4.06	5
	300	$\hat{K} = K$	99.3	99.9	100	100	99.5
		$\hat{K} > K$	0.7	0.1	0	0	0.5
		$\hat{K} < K$	0	0	0	0	0
		ave($\hat{K}$)	5.01	5	5	5	5
	500	$\hat{K} = K$	100	99.6	100	100	100
		$\hat{K} > K$	0	0.4	0	0	0
		$\hat{K} < K$	0	0	0	0	0
		ave($\hat{K}$)	5	5	5	5	5

NOTE: "ave($\hat{K}$)" is the average of the estimated number of common factors.

Table 5. Percentages of the estimated number of common factors for Case 5 with $n = 300$ in 1000 simulations: "$\hat{K} = K$," "$\hat{K} > K$," and "$\hat{K} < K$" truly estimates, overestimates, and underestimates the number of common factors, respectively.

			Gaussian population				
ρ	p		PC_3	ON_2	ER	GR	ACT
$\rho = 0.05$	100	$\hat{K} = K$	0	0	1.1	1.3	0.2
		$\hat{K} > K$	0	0	2.1	2.2	0
		$\hat{K} < K$	100	100	96.8	96.5	99.8
		ave($\hat{K}$)	1.29	1.75	1.57	1.59	3.29
	300	$\hat{K} = K$	38.4	26.7	5.1	5.6	54.5
		$\hat{K} > K$	0	0	0	0.1	0.8
		$\hat{K} < K$	61.6	73.3	94.9	94.3	44.7
		ave($\hat{K}$)	4.3	4.07	1.85	1.88	4.55
	500	$\hat{K} = K$	0.2	91.2	42.2	43.9	95.5
		$\hat{K} > K$	0	0	0	0	1.0
		$\hat{K} < K$	99.8	8.8	57.8	56.1	3.5
		ave($\hat{K}$)	2.68	4.91	3.04	3.11	4.97
$\rho = 0.15$	100	$\hat{K} = K$	0	0.1	0	0	70.4
		$\hat{K} > K$	0	0	0	0	0.5
		$\hat{K} < K$	100	99.9	100	100	29.1
		ave($\hat{K}$)	2.09	2.44	1.35	1.36	4.68
	300	$\hat{K} = K$	76.1	58.2	0	0	98.4
		$\hat{K} > K$	0	0	0	0	1.6
		$\hat{K} < K$	23.9	41.8	100	100	0
		ave($\hat{K}$)	4.76	4.55	1.83	1.84	5.02
	500	$\hat{K} = K$	2.0	99.8	0	0	99.1
		$\hat{K} > K$	0	0	0	0	0.9
		$\hat{K} < K$	98.0	0.2	100	100	0
		ave($\hat{K}$)	3.33	5	1.97	1.97	5.01
$\rho = 0.5$	100	$\hat{K} = K$	0	0.4	0	0	56.1
		$\hat{K} > K$	0	0	0	0	0
		$\hat{K} < K$	100	99.6	100	100	43.9
		ave($\hat{K}$)	2.23	2.54	1.99	2	4.5
	300	$\hat{K} = K$	78.2	60.2	0	0	100
		$\hat{K} > K$	0	0.1	0	0	0
		$\hat{K} < K$	21.8	39.7	100	100	0
		ave($\hat{K}$)	4.78	4.58	2	2	5
	500	$\hat{K} = K$	2.3	99.8	0	0	99.9
		$\hat{K} > K$	0	0	0	0	0.1
		$\hat{K} < K$	97.7	0.2	100	100	0
		ave($\hat{K}$)	3.35	5	2	2	5

NOTE: "ave($\hat{K}$)" is the average of the estimated number of common factors.

6.2. Transformed Macro Data After the Financial Crisis

The period covers the data from January 2010 to October 2018 with sample size of $n = 106$ and $p = 123$. Again, PC_2 selects nine factors. The nine largest eigenvalues are 5.68×10^4, 2.77, 0.86, 0.35, 0.17, 0.11, 0.08, 0.08, 0.03. Again, the marginal sample variances vary largely from 5.35×10^{-7} to 5.74×10^4. In contrast, our estimation method ACT chooses 7 factors. The nine largest eigenvalues of correlation matrix are 16.48, 12.20, 9.23, 5.75, 5.68, 5.27, 4.22, 3.82, and 3.53. Moreover, nine selected factors explain 53.83% total variation in 123 series, whereas 7 factors explain 47.85% total variation in 123 series, which is similar to the pre-crisis period by using the same method.

Example 2 (100 Fama–French portfolios). We now estimate the number of factors using the excess returns of Fama–French 100 portfolios. The data can be downloaded from the data library of Professor Kenneth French's website. Again, we divide the data into two periods: before and after the financial crisis.

6.3. Before the Financial Crisis

Following Fan, Zhang, and Yu (2012), we use the daily returns of 100 industrial portfolios formed on the basis of size and book-to-market ratio from January 2, 1998 to December 31, 2007. We note that the 71st and 100th portfolios have very large variances that possibly jeopardize the covariance matrix based methods. PC_3, IC_3, ON_2, ER, and GR estimate the number of factors as 10, 10, 6, 3, and 3, respectively. The largest 10 eigenvalues of the sample covariance matrix are 1824.45, 885.13, 117.39, 9.74, 5.38, 3.17, 2.31, 2.14, 1.86, 1.59. Ten factors explain 98.86% total variation in the 100 portfolios; four factors (suggested by ACT) explain 98.29% total variation in 100 portfolios. The largest 10 eigenvalues of sample correlation matrix are 65.81, 5.74, 2.57, 1.95, 1.10, 0.97, 0.90, 0.83, 0.72, 0.63. Ten factors explain

Table 6. Percent R^2 of well-known risk factors explained by PC-factors.

	Rm-Rf	SMB	HML	Momentum
Before crisis (4 selected factors)	0.953	0.931	0.829	0.141
Before crisis (3 selected factors)	0.947	0.813	0.821	0.132
After crisis (3 selected factors)	0.982	0.891	0.917	0.155

81.26% total variation, whereas four factors explain 76.09% total variation in 100 portfolios, in the standardized variables.

The well-known risk factors for equity markets are Fama–French factors (Fama and French 1993, 2015) and the momentum factor (Carhart 1997). To examine how these known factors can be explained by the unsupervised learning method (PCA with number of factors selected by ACT), we regress each known risk factor on the four principal component factors and report the coefficients of determination R^2 in Table 6. As comparison, we also regress these Fama–French factors on the three selected factors and report the results in the same table.

First of all, the well-known Fama–French factors (Rm-Rf is the market factor; SMB is the size factor; HML is the value factor; these three factors are nearly uncorrelated) are explained very well by the factors learned from the principal components. Regarding PC-factors as true factors (subject to learning or estimation errors), the results lend further support that the Fama–French factors are three most important factors, spanning essentially the same space as the first three PC-factors (regressing on first three PCs yields similar results). Such a confirmation of Fama–French factors appears new. On the other hand, the momentum factors cannot be explained well by the first four principal components, which is a surprise. Figure 1 depicts how well these four well-known equity risk factors can be explained by the four principal components. As expected, the four principal components explain the Fama–French factors better than the first three principal components do. On the other hand, the Carhart's momentum factor is not supported by the principal components.

We measure the difference between four given risk factors and four learned factors by using their projection matrix. Let A be a $n \times 4$ matrix formed by the time series of the four known factors and B be an $n \times 4$ matrix formed the four principal component factors. Define the projection matrix as $P_A = A(A^TA)^{-1}A^T$ and $P_B = B(B^TB)^{-1}B^T$. We then measure the difference between the space spanned by the four well-known factors and four learned PC-factors by using the Operator norm and Frobenius norm. For our data, they are

$$\|P_A - P_B\|_2 = 0.973, \quad \|P_A - P_B\|_F = 1.591.$$

In a similar vein, we measure the difference between the three Fama–French factors (A) and the three principal factors (B) by using the projection matrices. They are

$$\|P_A - P_B\|_2 = 0.481, \quad \|P_A - P_B\|_F = 0.949.$$

6.4. After the Financial Crisis

We extract the data from January 4, 2010 to April 30, 2019. The covariance matrix based methods PC_3, IC_3, ON_2, ER, and GR estimate number of factors as 6, 6, 6, 1, and 1, respectively. On the other hand, ACT selects three factors which explain 85.90% total variation in 100 portfolios with the three largest eigenvalues being 80.62, 3.22, and 2.06 based on the sample correlation matrix.

The R^2 of each the four well-known risk factors determined by three principal component factors is depicted in Table 6. Again, this confirms once more that the famous Fama–French factors aligned well with the first three principal components. Indeed, the differences between the two spaces are

$$\|P_A - P_B\|_2 = 0.406, \quad \|P_A - P_B\|_F = 0.708,$$

smaller than those before the financial crisis. On the other hand, the momentum factors are still not explained well by the PC factors.

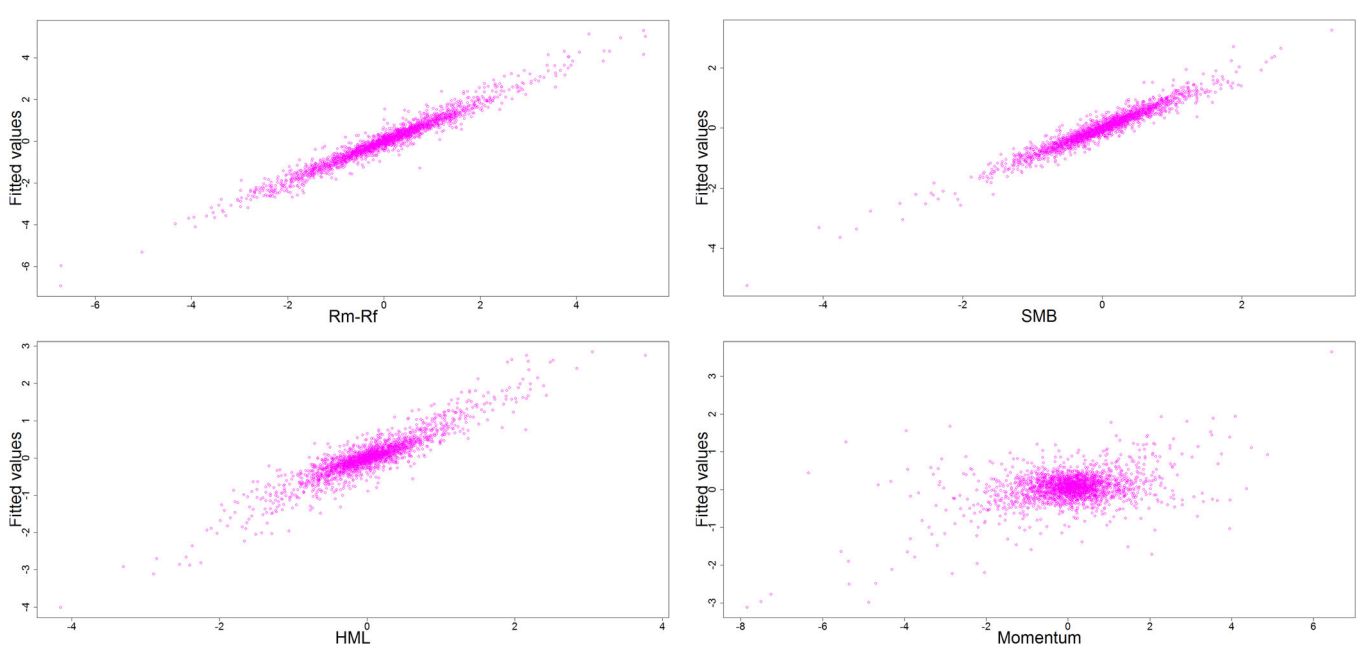

Figure 1. Graphical display for every observable factor versus its fitted values by regressing every observable factor on the four principal component factors before economic crisis.

7. Conclusions

Based on the correlation matrix, this article unveils the equality between the number of eigenvalues exceeding one and the number of latent factors. To use this relationship, we study random matrix theory based on the sample correlation to correct the biases in estimating the top eigenvalues and to take into account estimation errors in eigenvalue estimation. This naturally gives rise to the ACT for determining the number of common factors in high-dimensional factor models. The estimation method overcomes the disadvantages of using the sample covariance matrix which allows observable variables to be highly heterogeneous in scale. Simulation studies show that our proposed estimation method outperforms competing methods in the literature. This article considers the iid samples from the static factor model. But in practice, people also care about the dynamic factor model. Our future work will establish the relationship between the population correlation matrix and the number of common factors in the dynamic factor models, and propose estimating the number of factors in the high-dimensional dynamic factor model.

Supplementary Materials

Title: Supplementary materials for "Estimating Number of Factors by Adjusted Eigenvalues Thresholding." The supplementary materials include nine lemmas and their proofs, the proofs of Theorems 1–3 and simulation results for uniform population. (SuppleFileFactor.pdf)

R codes for ACT: R codes are utilized for simulation studies in Section 5 and empirical studies in Section 6 (simuexam zipped file).

Datasets: Datasets are utilized in empirical studies in Section 6. (data zipped file)

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